

1. Administrative & Study Identification

Full Study Title: A Study of Nemtabrutinib (MK-1026) Versus Comparator (Investigator's Choice of Ibrutinib or Acalabrutinib) in First Line (1L) Chronic Lymphocytic Leukemia (CLL)/ Small Lymphocytic Lymphoma (SLL) (MK-1026-011/BELLWAVE-011) **Official Title:** A Phase 3, Randomized Study to Compare Nemtabrutinib Versus Comparator (Investigator's Choice of Ibrutinib or Acalabrutinib) in Participants With Untreated Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma (BELLWAVE-011) **Short Title/ Acronym:** BELLWAVE-011 **Protocol Number:** MK-1026-011 **NCT Number:** NCT06136559 **Posting ID:** 174625704526148568 **Trial Status:** RECRUITING (Currently recruiting participants) **Sponsor/Company Name:** Merck **Investigational Product:** Nemtabrutinib (MK-1026) **Phase of Development:** PHASE3 (Trial must be in PHASE3) **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether nemtabrutinib is non-inferior to investigator's choice of ibrutinib or acalabrutinib with respect to Objective Response Rate (ORR), and whether nemtabrutinib is superior to ibrutinib or acalabrutinib with respect to Progression-Free Survival (PFS), as assessed by Blinded Independent Central Review (BICR) per International Workshop on Chronic Lymphocytic Leukemia (iwCLL) Criteria 2018. **Secondary Objectives:** To evaluate Overall Survival (OS), Duration of Response (DOR) per iwCLL Criteria 2018 by BICR, and the number of participants who experience one or more Adverse Events (AEs).

2.2 Background & Rationale To address the unmet medical need in previously untreated Chronic Lymphocytic Leukemia (CLL) and Small Lymphocytic Lymphoma (SLL). Current treatments involving covalent Bruton tyrosine kinase (BTK) inhibitors (like ibrutinib and acalabrutinib) have improved survival but are often associated with substantial toxicities, drug resistance (e.g., C481 mutations), and low complete remission rates. This study evaluates if nemtabrutinib—a reversible, non-covalent inhibitor of wild-type and mutated BTK—offers better efficacy outcomes and a manageable safety profile in first-line (1L) treatment.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Investigator's choice of Ibrutinib or

Acalabrutinib) **Blinding:** Open-label **Randomization:** 1:1, Stratified

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 1,200 participants **Number of Sites:** Global, Multicenter **Estimated Study Start:** 14-Dec-2023 **Estimated Primary Completion:** 23-Nov-2032 (approximate total duration up to 9 years)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Confirmed diagnosis of CLL/SLL and active disease clearly documented to have a need to initiate therapy. 2. Has at least 1 marker of disease burden. 3. Has an Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 2 within 7 days before randomization. 4. Has the ability to swallow and retain oral medication. 5. Participants who are hepatitis B surface antigen (HBsAg) positive are eligible if they have received hepatitis B virus (HBV) antiviral therapy for at least 4 weeks and have undetectable HBV deoxyribonucleic acid (DNA) viral load before randomization. 6. Participants with a history of hepatitis C virus (HCV) infection are eligible if HCV ribonucleic acid (RNA) viral load is undetectable at screening. 7. Participants with human immunodeficiency virus (HIV) who meet ALL eligibility criteria.

4.2 Exclusion Criteria 1. Has an active hepatitis B virus/hepatitis C virus (HBV/HCV) infection. 2. Has gastrointestinal (GI) dysfunction that may affect drug absorption. 3. Has a diagnosis of Richter Transformation or active central nervous system (CNS) involvement by CLL/SLL. 4. Has had acquired immune deficiency syndrome (AIDS)-defining opportunistic infection in the past 12 months before screening. 5. Has clinically significant cardiovascular disease. 6. Has hypersensitivity to nemtabrutinib or contraindication to ibrutinib or acalabrutinib, or any of the excipients. 7. Has a history of severe bleeding disorder. 8. Has known additional malignancy that is progressing or has required active treatment within the past 2 years. 9. Has received any systemic anticancer therapy for CLL/SLL. 10. Is currently being treated with p-glycoprotein (P-gp) substrates with a narrow therapeutic index, cytochrome P450 3A (CYP3A) strong or moderate inducers or CYP3A strong inhibitors. 11. Received prior radiotherapy within 2 weeks of start of study intervention, or radiation-related toxicities, requiring corticosteroids. 12. Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention. Administration of killed vaccines is allowed. 13. Has received an investigational agent or has used an investigational device within 4 weeks before study intervention administration. 14. Has an active infection requiring systemic therapy, including intravenous (IV) antibiotics during screening. 15. Participants who have not adequately recovered from major surgery or have ongoing surgical complications.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Investigational Arm:** Nemtabrutinib 65 mg once a day (QD). * **Comparator Arm:** Ibrutinib (Trade Name: Imbruvica | ATC Code: L01EL01 | [EMA Product Info](#)) 420 mg once a day (QD) OR Acalabrutinib 100 mg twice a day (BID). **Route of Administration:** Oral **Duration of Treatment:** Continuous treatment until unacceptable toxicity, progressive disease, or withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care measures as indicated by the investigator. Killed vaccines. **Prohibited:** Other concurrent investigational agents, devices, or anti-cancer therapies for CLL/SLL. Live or live-attenuated vaccines. P-gp substrates with a narrow therapeutic index, CYP3A strong/moderate inducers, and CYP3A strong inhibitors.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Physical Exam, Labs, Disease Burden Assessment
Baseline	Day 0	Randomization, First Dose Administration
Interim	Q12W (up to Wk 97)	Safety Labs, Efficacy Assessment (Imaging, Blood/Marrow assessment), AE monitoring
Long-Term	Q24W (post Wk 97)	Continued Efficacy and Safety Assessments
End of Study	At Discontinuation	Final Physical Exam, Exit Interview, Study Drug Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Objective Response Rate (ORR)** per iwCLL Criteria 2018 as assessed by BICR. Defined as the percentage of participants with complete response (CR), complete response with an incomplete recovery of the participant's bone marrow (CRi), nodular partial response (nPR), or partial response (PR). 2. **Progression-Free Survival (PFS)** per iwCLL Criteria 2018 as assessed by BICR. Defined as the time from randomization to the first documented progressive disease (PD) or death due to any cause, whichever occurs first.

7.2 Secondary Outcomes 1. **Overall Survival (OS)** 2. **Duration of Response (DOR)** per iwCLL Criteria 2018 as assessed by BICR 3. **Number of Participants Who Experience One or More Adverse Events (AEs)**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring through physical exams and laboratory parameters to track the number of participants experiencing AEs. **Serious Adverse Events (SAEs):** Reported to the sponsor typically within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** External independent Data Monitoring Committee to evaluate ongoing safety and efficacy parameters.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary and secondary efficacy analyses; Safety Set for safety/AE summaries. **Sample Size Justification:** Target enrollment of $N = 1,200$ is adequately powered to test the primary hypotheses (non-inferiority of ORR and superiority of PFS). **Handling of Missing Data:** Efficacy endpoints like PFS and OS will use standard time-to-event right-censoring methods.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence and data integrity. **Audit Trail:** Secure eCRF entry, full audit trails, and data clarification mechanisms implemented by the sponsor.

11. Study Identifiers & Governance

Primary Protocol ID: MK-1026-011 **Registry Number:** NCT06136559
Posting ID: 174625704526148568 **Sponsor/Company:** Merck **Study Title:** BELLWAVE-011 **Trial Status:** RECRUITING (Currently recruiting participants) **Phase:** PHASE3

12. Clinical Framework (Ontology & Disease Specifics)

Primary Conditions: Chronic Lymphocytic Leukemia (CLL), Small Lymphocytic Lymphoma (SLL) **Disease Taxonomy/Ontology Details:** * **Name:** Leukemia (Preferred Name: leukemia) * **Semantic Type:** Neoplastic Process * **Definition:** A cancer of the blood and bone marrow characterized by an abnormal proliferation of leukocytes. [<https://orcid.org/0000-0002-0736-9199>, NCIT:C3161] * **Ontology Codes:** MeSH Code: D007938 | HPO Code: HP:0001909 | SNOMED ID: 93143009 * **SNOMED Hierarchy Base:** Malignant neoplasm of lymphoid, hematopoietic and/or related tissue (disorder) **Medical Methodology:** BTK inhibition (reversible, non-covalent vs.

covalent) **Optimization Target:** Disease remission (ORR) and extended Progression-Free Survival (PFS)

13. Study Procedures & Methodology

Management Pathway: Targeted continuous oral therapy until disease progression **Education Protocol (HCP):** Site initiation visits and comprehensive protocol/AE management training **Education Protocol (Patient):** Informed consent process, dosing diaries, and self-monitoring guidance for oral therapies **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for primary efficacy confirmation per iwCLL 2018 criteria

14. Observation & Follow-Up Schedule

Total Duration: Up to 9 years (until progression or unacceptable toxicity) **Onsite Visit Cadence:** Every 12 weeks (Q12W) up to week 97; Every 24 weeks (Q24W) thereafter **Remote Monitoring Frequency:** Periodic virtual safety check-ins as defined by the protocol **Checklist Review Interval:** Reviewed at each scheduled clinic visit

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]					
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]					